

Injection Laser Diodes

PHYS3117 Experimental Physics

Student Notes

1 Experimental Aim

The aim of this experiment is to gain an understanding of the operating characteristics of some semiconductor diode lasers. The output power versus injection current and the output optical frequency spectra are examined using a power meter and a Fabry-Perot interferometer.

1.1 Learning Outcomes

- Understand the operating principles of a diode laser in terms of the microscopic theory of semiconductors and basic laser theory.
- Understand the physical optics principle responsible for strongly asymmetric and divergent beam profile observed in edge-emitting diode lasers
- Learn how to perform current-power measurement on different laser diodes and assess device characteristics such as slope-efficiency and threshold current
- Understand the effect of temperature tuning of diode lasers device properties in terms of solid-state physics.
- Understand the operation of a Fabry-Perot interferometer and how it can act as a spectrometer.

2 Introduction

Semiconductor diode lasers are devices that are used for many common applications and in many instruments. A laser diode is a critical component in laser printers, CD players, DVDs, CD-ROM read and recording devices and bar code readers. They are also used as sources for optical communications systems and as pump sources for solid-state lasers. They are usually employed because they are relatively inexpensive (compared with other laser types), are reliable, small in size, easy to use and require only low powers.

The operating principles of a diode laser are similar to any other laser system. The device consists of a gain medium with feedback. Early diode lasers consisted of a forward biased (homo-structure) GaAs p-n junction. Optical gain was achieved via electron-hole recombination in the depletion region and feedback was achieved via Fresnel reflection from the cleaved end facets. Today, many more complicated types of diode laser exist with a variety of internal structures such as hetero-structure, double hetero-structure, multi quantum-well, and vertical cavity surface emitting). They are fabricated from various materials such as InAs, InP, GaAsP, GaInAs, and InAsP and are capable of operating at wavelengths from the near ultraviolet to the far infrared.

A temperature control mount is used with one of the diode lasers to demonstrate the relationship between temperature and output power. In many applications, precise control of the temperature is necessary to obtain stable output with desired characteristics. Temperature control is particularly important for high power devices (up to several watts) where active cooling is required so that the diode lasers do not overheat.

Semiconductor diode lasers have unique spectral characteristics that arise due to their small size. A scanning Fabry-Perot Interferometer (FPI) is a device commonly used for examining laser spectral features. It consists of two plane parallel mirrors coated for high reflectivity at the appropriate wavelengths. One mirror position is driven by an oscillating ramp signal, which provides frequency discrimination. Important characteristics of FPIs include the “free spectral range” and the “finesse”.

3 Experimental Instructions

3.1 Precautions

Three different light sources are used in this experiment: two low power visible diode laser (class 3R), and a higher power visible diode laser (class 3B). Due to the small beam cross section, each of these devices is capable of delivering power densities which may permanently damage the retina of the eye if the laser beam (or a specular reflection) is viewed directly. To avoid the possibility of eye damage the following should be observed:

1. Before undertaking any realignment of the laser diode beampath, ensure that you are wearing the appropriate laser safety goggles.
2. Never view the laser beam (or a specular reflection) either directly or through any optical instrument. Use a photo-detector, white card, or infrared viewing card to detect the beam.
3. Remove any rings, watches, etc., and keep the working bench clear of unnecessary apparatus to minimise the chance of strong stray reflections.
4. Do not endanger other people's eyes. Make sure the main beam and any strong reflections are terminated on your experimental bench. Black screens are available for this purpose.

We will be three different lasers for these experiments:

1. Diode Laser A (LDA): DL3148-025: **current limit: 45 mA**
2. Diode Laser B (LDB): HL6322G: **current limit: 100 mA**
3. Diode Laser C (LDC): ML 1016R MQW: **current limit: 70 mA**

LDA and LDB use the OPC-PS03-A diode laser driver and LDC uses the SDL 800 diode laser driver. Ensure that the current limit is for each laser prior to enabling the diode driver.

3.2 Spatial characteristics of an edge-emitting diode laser

1. Connect the DL3148-025 diode laser (labeled - LDA) to the laser.
2. Observe the laser output beam when the injection current is set to $\sim 50\text{ mA}$ on a white card and measure the beam divergence in both directions. Note: there is no collimating lens in front of the LDA output.
3. Using basic diffraction arguments, estimate cross-sectional dimensions of the laser.

Question 1: How do the spatial characteristics of diode laser compare with other laser types e.g. HeNe? Comment.

3.3 MQW diode laser output characteristics

A typical heterostructure diode laser has an active layer width of $\sim 0.2\text{ }\mu\text{m}$. Reducing the active layer thickness reduces the volume which has to be electrically pumped, resulting in the same gain for a far lower injected current density. Thus, for higher output power and lower threshold current, the layer width should be as small as possible. Additionally, improved spatial mode confinement is achieved for the narrowest layer thickness. The active layer of a multi quantum-well (MQW) diode laser consists of a series of ultra-thin ($\sim 0.2\text{ }\mu\text{m}$) layers composed of alternating materials, in this case InGaP and InGaAlP on a GaAs substrate.

1. Connect the HL6322G laser diode (LDB) into the diode laser driver. This laser has an aspheric lens installed to improve the collimation. Turn on the diode laser following the diode laser driver operating procedure above (AFTER YOU HAVE SPOKEN WITH A DEMONSTRATOR). Turn the injection current to $\sim 50\text{ mA}$.
2. Use the fibre coupled spectrometer (Ocean Optics, Flame) to record the spectrum at a number of different injection currents above and below the threshold current. You may be required to vary the integration time or the position of the fibre in order to avoid saturating the CCD in the spectrometer or improve the signal-to-noise. Explain the differences in the observed emission spectrum above and below threshold.
3. Place a power meter (with attenuator) as close as possible to the diode laser front facet. Ensure that the power meter is calibrated to the correct wavelength (635 nm). Plot the output power as a function of injection current, but do not exceed 10 mW

output power. Determine the threshold injection current, I_{th} , and slope efficiency, dP/dI . Compute the overall efficiency:

$$\eta = \frac{dP}{dI}(\lambda[\mu m]/1.24) \quad (1)$$

How do these values compare with the specifications?

Question 2: Explain what is meant by the laser threshold gain condition.

Question 3: What factors determine the threshold injection current for a diode laser?

3.4 Temperature tuning of a diode laser

In this experiment we aim to measure the change in threshold current and slope efficiency as a function of diode laser temperature, for one particular diode laser.

As the temperature of a diode laser is increased the threshold current density and the external differential quantum efficiency also increase. The characteristic temperature, T_0 of a diode laser is a parameter used to describe the temperature sensitivity. It is related to the change in threshold injection current, I_{th} with temperature, T by:

$$T_0 = \frac{T}{\ln(I_{th})} \quad (2)$$

In this section a temperature control mount is used to adjust the operating temperature of the diode laser. This mount consists of a Peltier heating/cooling element, a thermocouple to monitor the diode temperature and a heat sink to dissipate the heat. Both cooling and heating can be accomplished by changing the polarity of the Peltier element

1. Ensure that the the ML1016R diode laser (LDC) is plugged into the the diode laser driver SDL 800. Set the SETPOINT temperature to 23 °C. Observe the ACTUAL TEMPERATURE and wait until it has stabilised (~ 5 minutes).
2. Turn on the laser diode and set the injection current to 40 mA. Position the power meter about 10 mm from the front of the laser diode lens. As before, measure the output power versus injection current and determine the diode laser threshold injection current.
3. Turn the diode laser injection current down and switch to SETUP mode. Change the SETPOINT temperature to 20 °C. Observe the ACTUAL temperature and wait until the temperature is stabilised (~10 minutes). Turn the laser ON and re-measure the threshold injection current. Again do not exceed 3 mW output power. Repeat the procedure for another few temperatures down to 13 °C.
4. Plot threshold injection current versus temperature, and determine the characteristic temperature of this diode laser.
5. Replace the photodiode with the USB spectrometer and set the injection current to 50 mA. Measure the emission spectrum as a function of temperature over the same

temperature range. Ensure that the temperature has stabilised before recording the spectrum. Plot the peak wavelength as a function of temperature and comment on the results.

Question 4: Describe and explain the relationship you observe between threshold injection current and temperature.

Question 5: What factors contribute to the output power available in a diode laser? Why is higher output available from this diode laser compared with diode laser LDB?

3.5 Spectral characteristics of a diode laser

The aim of this next section is to examine the output optical frequency spectrum of a diode laser, using a Fabry-Pérot interferometer (FPI).

The Fabry-Pérot interferometer is a device consisting of two plane parallel mirrors partially transmitting at visible wavelengths. Constructive interference occurs when the spacing between the mirrors is an integral number of half wavelengths. The Fabry-Pérot controller sends a ramped voltage to the mirrors, which oscillates the mirror spacing via a piezoelectric element. The photodiode then amplifies the optical signal, generating the optical frequency spectrum of the diode laser on the oscilloscope. Modes will be discriminated only if the mirrors are exactly parallel and the beam is exactly perpendicular to the mirror surfaces.

3.5.1 Procedure

WARNING: the FPI mirrors are very close together (to achieve a large free spectral range) so that great care must be taken not to scratch them against one another. THESE MIRRORS ARE VERY EXPENSIVE!! Therefore, only turn the mirror adjustment knobs by a very small amount each time. Ask a demonstrator if you are not sure.

1. Ensure that the the ML1016R diode laser (LDC) is plugged into the the diode laser driver SDL 800. Set the SETPOINT temperature to 23 °C. Observe the ACTUAL TEMPERATURE and wait until it has stabilised (~ 5 minutes) Turn the diode injection current to a level that is above the threshold current.
2. Before you attempt to align the Fabry-Perot interferometer consult a demonstrator who will instruct you in its use. *The FPI can be a difficult instrument to align properly.* Make sure you understand the principles of its operation before using it.
3. The LDC laser is positioned such that no additional steering mirrors should be necessary to align the output to the Fabry-Pérot Interferometer. Small adjustments in the Fabry-Pérot position and orientation are required to direct the back-reflection from the first mirror back to the laser diode aperture.
4. Place a white card in front of the photodiode and observe the laser spot. The second FP mirror must now be adjusted such that the mirrors are perfectly parallel to one another. When this occurs the laser beam will be a point source (spot) rather than

- a line. This is an iterative process. Each of the two large mirror adjustment screws (large knobs) should be turned one after the other until the best point source is observed (you should also see interference fringes). Do not adjust the third (and lowest) smaller mirror adjustment screw.
5. Place a white card in front of the photodiode and observe the laser spot. The second FP mirror must now be adjusted such that the mirrors are perfectly parallel to one another. When this occurs the laser beam will be a point source (spot) rather than a line. This is an iterative process. Each of the two large mirror adjustment screws (large knobs) should be turned one after the other until the best point source is observed (you should also see interference fringes). Do not adjust the third (and lowest) smaller mirror adjustment screw.
 6. Turn on the FPI ramp driver. Set the ramp duration to 50 ms, the ramp amplitude to about half maximum and the magnification to $\times 1$.
 7. Turn on the photodiode amplifier. Observe the output of the photodiode on channel 1 of the oscilloscope (1V/div, 10 ms/div, Invert = on). You will need to turn off the laboratory lights so that the photodiode does not saturate. You may also need to place a black cloth over the photodiode. Observe the ramp signal (output/100) from the back of the FPI ramp driver on channel 2 of the oscilloscope (5V/div). Use the trigger signal, from the back of the FPI ramp driver, as the External Trigger of the oscilloscope. You will also need to insert one or more neutral density filters (3.0, 4.0 or 0.3, or a combination of these) in front of the photodiode so that the signal is not saturated.
 8. Now do a fine adjustment on all of the FPI positioning screws, the FPI mirror adjustment knobs to achieve the highest finesse i.e. so that the lasing mode(s) are as thin and as high as possible.
 9. When you have the best defined laser mode profile, capture the data from the oscilloscope. Measure the separation between laser modes using the oscilloscope cursor.
 10. Assuming that the FSR is 1000 GHz, calculate the length of the laser diode cavity.
 11. Examine the behaviour of the mode structure as the injection current is reduced close to the threshold value. Comment on your results.
 12. Now turn the laser diode injection current to zero and turn off the diode laser driver.

Question 6: Explain the term gain bandwidth.

Question 7: What factors determine the operating wavelength of a diode laser?

Question 8: What is the difference between an LED and a diode laser?

Question 9: What is a longitudinal mode and what determines the longitudinal mode separation?

Question 10: The semiconductor gain profile is largely homogeneously broadened, yet side modes carry significant power. Why?

Question 11: At this Fabry-Perot FSR (=1000 GHz), what can be said about the linewidth of each longitudinal mode?

Question 12: Discuss the differences operation of the USB grating spectrometer and the FPI spectrometer.